<u>Notes</u>: 1) Approximate standard error of $\bar{x}$ is

$$\hat{se}(\bar{x}) = \frac{s}{\sqrt{n}}$$

2) Approximate margin of error at $(1-\alpha)100\%$ ~~confi~~ confidence is

$$\widehat{MOE} = t_{\frac{\alpha}{2}, n-1} \frac{s}{\sqrt{n}}.$$

3) Approximate width/length of CI

$$\hat{L} = 2 \times \widehat{MOE} = 2 \cdot t_{\frac{\alpha}{2}, n-1} \frac{s}{\sqrt{n}}.$$

<u>Theorem</u>: Approximate CI for μ (σ-unknown) for <u>non-normal</u> distribution/population.

Suppose $X_1 \ldots X_n$ is a RS from a non-normal distribution/population with μ, σ^2 both unknown.

<u>If $n \geq 30$</u>, then an approximate $(1-\alpha)100\%$ CI for μ is

$$\bar{x} \pm t_{\frac{\alpha}{2}, n-1} \frac{s}{\sqrt{n}}.$$

Hypothesis test for μ. (σ known)

3 - step test.

$H_0: \mu = \mu_0$

↑ null hypo

↑ value.

alternate hypo.

$H_a: \mu \neq \mu_0$ two-sided test

$H_a: \mu > \mu_0$ right-sided test

$H_a: \mu < \mu_0$ left-sided test.

1) Get test statistic.

$$Z^* = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}.$$

2) critical value. (based on α) for example $\alpha = 0.05$.

two-sided test: $Z_{\frac{\alpha}{2}} = Z_{0.025} = 1.96$

right-sided test: $Z_\alpha = Z_{0.05} = 1.645$

left-sided test: $-Z_\alpha = -Z_{0.05} = -1.645$

3) Decision and final conclusion.

two-sided test: i) If $|Z^*| > Z_{\frac{\alpha}{2}}$, then reject H_0

We have strong enough evidence to support H_a.

ii) If $|Z^*| \leq Z_{\frac{\alpha}{2}}$, then fail to reject H_0

We don't have strong enough evidence to support H_a.

right-sided test : If $Z^* > Z_\alpha$, then reject H_0.

We have strong enough evidence to support H_a.

left-sided test : If $Z^* < -Z_\alpha$, then reject H_0

We have strong enough evidence to support H_a

<u>Example</u> : Bags of sugar have a label weight, which is 16 oz.

The weights, X, can be modeled by a <u>normal distribution</u> with mean μ and var $\sigma^2 = 0.05^2$. $\alpha = 0.05$.

Hypothesis test for $H_0 : \mu = \underbrace{16}_{\mu_0}$ vs. $H_a : \mu \neq 16$.

An inspector R.S. <u>25</u> bags, and get $\bar{x} = 15.974$ oz.

1) Get statistic

$$Z^* = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}} = \frac{15.974 - 16}{0.05 / \sqrt{25}} = -2.60$$

2) critical value.

$$Z_{\frac{\alpha}{2}} = Z_{0.025} = 1.96$$

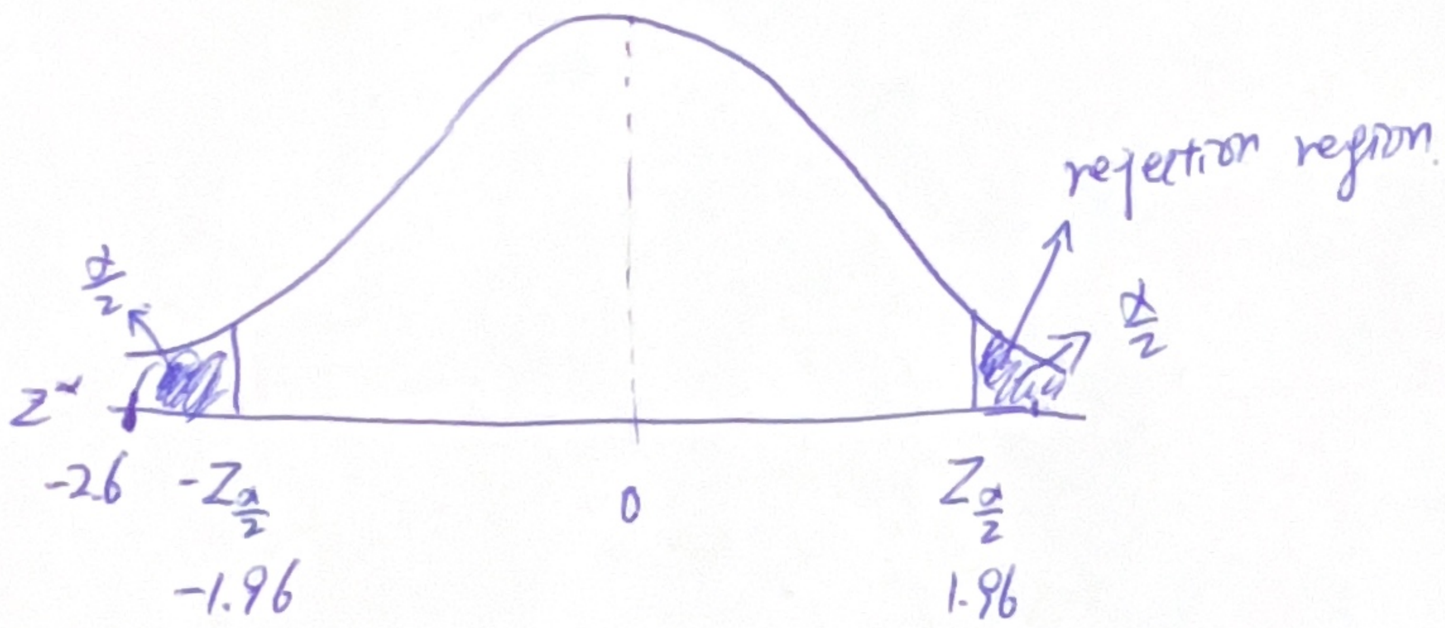
3) Decision and final conclusion.

$$|Z^*| = 2.60 > 1.96 = Z_{\frac{\alpha}{2}}.$$

Therefore reject $H_0 : \mu = 16$. We have strong enough evidence to support the population mean of supar bags not equal to ~~16~~ 16 oz

μ significantly differs from 16 oz.

* How to understand the process?



P-value

Def'n: P-value = P(get a test statistic at least as extreme as you just observed | H_0 is true)

two-sided: p-value = $2P(Z > |Z^*|)$

right-sided: p-value = $P(Z > Z^*)$

left-sided: p-value = $P(Z < Z^*)$

Rule: If p-value $< \alpha$, reject H_0

If p-value $\geq \alpha$, fail to reject H_0.